Practice assignment for Lesson 2.3.4

Practice Assignment • 10 min

← Back

English

Your grade: 100%

Your latest: **100%** • Your highest: **100%**

To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the strategy presented in this lesson for implementing a Kalman filter for smoothing? — 1 / 1 point

- ○ Start with the output of a standard Kalman filter and add steps to provide an estimate of the states of the system at a future time.
- ◉ Add steps to the standard Kalman filter to compute a smoothed past state estimate based on a saved trajectory of state estimates from the standard Kalman filter.
- ○ Rederive the Kalman-filter algorithm, developing new steps 1a-2c to produce the smoothed past states.
- ○ Simply use the output of the standard Kalman filter without changes.

Correct
Yes. That is correct.

2. Which of the following is true for the results presented in this lesson? — 1 / 1 point

- ○ Error bounds on smoothed states are wider than those for estimated states.
- ◉ Error bounds on smoothed states are narrower than those for estimated states.
- ○ Error bounds on smoothed states are the same size as those for estimated states.
- ○ Error bounds on smoothed states may either be greater or less than those for estimated states.

Correct
Yes. This is correct.

3. Execute the Coursera Labs Jupyter notebook for this lesson. What is the root-mean-squared position error for the standard Kalman filter? Enter your answer with three digits to the right of the decimal point. — 1 / 1 point

0.003

Correct
Yes. That is correct.

4. Execute the Coursera Labs Jupyter notebook for this lesson. What is the root-mean-squared position error for the Kalman smoother? Enter your answer with three digits to the right of the decimal point. — 1 / 1 point

0.002

Correct
Yes. That is correct.